median or where it is advantageous to widen the median on a tangent alignment, the change should be effected by reverse curves of one (1) degree or less.

Climbing lanes on multilane roads may be justified where the critical length of grade is exceeded and the reduced design capacity because of trucks climbing long grades is exceeded by the Directional Design Hourly Volume (DDHV) 30% or more. Geometric details are the same as for climbing lanes on 2-lane highways.

In the design of divided highways the inclusion of a median in the cross section alters somewhat the superelevation run-off treatment. The following are the different cases in attaining the superelevation runoff treatment:

- The whole of the travelled way, including the median, is superelevated as a plane section.
- The median is held in a horizontal plane and the two pavements are rotated about the median edges.
- The two pavements are separately treated for runoff, resulting in variable difference in elevation at the median edges.

1.5.5 Classification of Highways According to System

In the Philippines roads or highways are classified as national, provincial, city, municipal or barangay roads.

1.5.5.1 National Roads

Public roads, declared as national roads by the President of the Philippines upon recommendation of the Secretary of Public Works and Highways satisfying the conditions set forth under Executive Order No.113, Establishing the Classification of Roads. National roads are classified as primary and secondary roads. The former forms the part of the main highway trunk-line system which is continuous in extent; the latter includes all access roads forming a secondary trunk-line system.

- Road Right of way……..20.00 m minimum
- Width of travelled way 2 lane ……..6.70 m minimum
- Allowable grade ……6.0% maximum

1.5.5.2 Provincial Roads

These are roads connecting one municipality to another, with the terminal to be the public plaza; plus roads extending from one municipality or from a provincial or national road to a public wharf or railway station. For purposes of allocating national aid maintenance, a provincial road is designated and accepted as such by the Secretary of the Department of Public Works and Highways, upon recommendation of the Provincial Board (Sangguniang Panlalawigan).

- Road Right of way……15.00 m minimum
- Width of travelled way……6.10 m minimum
- Allowable grade……6.0% maximum

1.5.5.3 City Roads

Roads / streets within the urban area of a city not classified as provincial or national roads.

- Road Right of way ..... 15.00 m minimum
- Width of travelled way ..... 6.10 m minimum
- Allowable grade ..... 6.0% maximum

1.5.5.4 Municipal Roads

Roads / streets within the poblacion area of a municipality not classified as provincial or national roads.

- Road Right of way ..... 15.00 m minimum
- Width of travelled way ..... 6.10 m minimum
- Allowable grade ..... 6.0% maximum

1.5.5.5 Barangay Roads

Roads located outside the poblacion area of a municipality or urban area of a city and those outside industrial, commercial or residential subdivision (access roads to subdivisions are not barangay roads), and which act as a feeder from Farm-to-market road, and are not otherwise classified as national, provincial, city or municipal roads. Barangay roads must meet the following criteria:

- Road Right of way ..... 10.00 m minimum
- Width of travelled way ..... 4.00 m minimum
- Allowable grade ..... 10.0% maximum

1.5.5.6 Tourism Road

Tourism road is a road which marketed as particularly suited for tourist. Tourist road may be formed when existing road are promoted with traffic sign and advertising material. Some tourist road such as Ternate-Nasugbu road are built for tourism purposes. Others maybe roadways enjoyed by local citizen in areas of unique or exceptional natural beauty. It is often developed because it promises to generate employment, enhance community infrastructure and assist in revitalizing the flagging economies in rural areas.

- Road Right of way ..... 2.10 m minimum
- Width of travelled way ..... 6.10 m minimum
- Allowable grade ..... 6.0 % maximum

1.5.5.7 Farm to Market Road

Farm to Market Roads refer to roads linking the agriculture and fisheries production sites, coastal landing points and post-harvest facilities to the market and arterial roads and highways.

- Road Right of way ..... 6.00 m minimum
- Width of travelled way ..... 4.00 m minimum
- Allowable grade ..... 10.0% maximum

1.5.6 Road Classification According to Primary Function

Road classifications based on the primary functions are the following:

1.5.6.1 Expressways

These are divided arterial highways for through traffic, with full or partial control of access and generally with grade separations at major intersections.

1.5.6.2 Parkways

Parkways are arterial highways for non-commercial traffic with full or partial control of access, usually located within a park or a ribbon of park-like development.

3 Geometric Design of Road Links

3.1 Introduction

Standards have been developed as a guide in the design of highways and to ensure that motorist's convenience, environmental safety, and aesthetic considerations are implemented in the most economical manner consistent with highway service considerations.

3.1.1 Departure from Standards

Design policies and standards generally represent minimum values. Higher standards may be used within reasonable economic limits. To ensure uniform practice, lower design standards may not be used without approval from the DPWH Office of the Secretary or the Bureau of Design.

3.2 Requirements for Design Analysis in Operating Environment

Highway design involves geometrically combining the elements which establish the road layout, that is, horizontal and vertical alignment, width of pavement and formation, cross slopes, etc., to ensure that the finished road will be an appropriate component of the traffic system. The system must be satisfactory for all relevant environmental conditions so that the highway could provide maximum service to the public with minimum hazard at reasonable cost.

It is necessary that it be constructed to endure and to provide adequate safe passage of vehicles. To achieve this objective the design must adopt certain criteria or standards for strength and uniformity.

These criteria and standards are subject to modifications since roads are intimately associated with environmental conditions, vehicles, human factors and economic considerations which seldom conform to mathematical concepts.

Table 3-1 and Table 3-2 contain the minimum design standards for Philippine highways. Standard cross-sections ae provided in Figure 3-1 to Figure 3-6.

Table 3-3, Table 3-4 and Table 3-5 provide AASHTO minimum recommended widths of traveled way and shoulders for local rural roads, rural collector roads, and rural arterials.

The width of an urban collector street will be the sum of the widths of lanes for moving traffic as determined from a capacity analysis, plus space for parking and bicycles, and including the median width where applicable. Lanes within the traveled way should range in width from 3.0 to 3.6 m.

Expressways should have a minimum of two 3.6 m wide through-traffic lanes for each direction of travel. Paved shoulders should be continuous on both the right and left sides of all expressway facilities.

Table 3-1 Minimum Design Standards for Philippine Highways – excluding Tourism Roads

Average Daily Traffic (ADT)	FMR	BARANGAY	SECONDARY		PRIMARY		EXPRESSWAY	
	Under 200	200-400	400 – 1000		1000 – 2000		More Than 2000	
			Minimum	Desirable	Minimum	Desirable	Minimum	Desirable
DESIGN SPEED (kph)								
Flat Topography	60	70	70	90	80	95	90	100
Rolling Topography	40	50	60	80	60	80	70	90
Mountainous Topography	30	40	40	50	50	60	60	70
RADIUS (meters)								
Flat Topography	120	160	160	280	220	320	260	350
Rolling Topography	55	85	120	220	120	220	160	280
Mountainous Topography	30	50	50	80	80	120	100	160
GRADE (percent)								
Flat Topography	6.0	6.0	5.0	3.0	4.0	3.0	4.0	3.0
Rolling Topography	8.0	7.0	6.0	5.0	5.0	5.0	5.0	4.0
Mountainous Topography	10.0	9.0	8.0	6.0	7.0	6.0	7.0	5.0
Pavement Width (meters)	4.0	5.5 – 6.0		6.10		6.70	6.70	7.30
Shoulder Width (meters)	0.5	1.0	1.50	2.00	2.50	3.00		3.00
Right of Way Width (meters)	20	30		30	30	30		60
Superelevation (meters/meter)	0.08 (max.)		0.08 (max.)		0.08 (max.)		0.08 (max.)	
NON PASSING SIGHT DISTANCE (meters)								
Flat Topography	70	90	90	135	115	150	135	160
Rolling Topography	40	60	70	115	70	115	90	135
Mountainous Topography	40	40	40	60	60	70	70	90
PASSING SIGHT DISTANCE (meters)								
Flat Topography	420	490	490	615	560	645	615	675
Rolling Topography	270	350	420	560	420	560	490	615
Mountainous Topography	190	270	270	350	360	420	420	490
Type of Surfacing	Gravel, crushed gravel or crushed stone, bitumen preservative treatment, single or double bituminous surface treatment,		Bituminous dense or open graded plant mix surface course, bituminous concrete surface course		Bituminous concrete surface course		Bituminous concrete surface course, Portland cement concrete pavement	

Source: Table 3.2 in Ministry of Public Works and Highways, 1984, Design Guidelines Criteria and Standards Volume II, Bureau of Design, Manila.

Table 3-2 Minimum Design Standards for Tourism Roads

Design Element	Requirement
Pavement Type	Portland Cement Concrete Pavement (PCCP) or Asphalt Concrete Pavement (ACP)
Pavement Width	Minimum of 6.1 m for two lanes
Pavement Thickness	230 mm for Portland Cement Concrete Pavement (PCCP)
	50 mm for Asphalt Concrete Pavement (ACP)
Shoulder Width Material	 Minimum of 1.5 m Minimum gravel surfacing
Roadway Cross Slope	1.5% for Portland Cement Concrete Pavement (PCCP)/ 2% ACP
Radius of Horizontal Curve	Minimum of 50 m
Length of Tangent between Point of Curvature (PC) and Point of Tangency (PT) of reverse curve	Minimum length of 30m
Length of Vertical Curve	Minimum length of 60 m
Design Speed	Minimum of 60 kph, 40 kph and 30 kph for flat, rolling and mountainous terrain respectively
Longitudinal Grade	Minimum of 0.50% on cut section and maximum of 12%
Side Cut Slope Ratio (H:V)	Cut slope of 1.5:1 to 1:1 for common materials Cut slope of 0.5:1 to 1:1 for soft/rippable rock Cut slope of 0.25:1 for hard/solid rock or natural break
Road Drainage	Box culvert: 25-year flood with sufficient freeboard to contain the 50-year flood Pipe culvert: 15-year flood with sufficient freeboard to contain the 25-year flood; minimum size of 910 mm in diameter
Slope protection	As needed
Road Safety Devices including Pavement Marking	Refer to DPWH Highway Safety Design Standards, Part 2 (May 2012)
Minimum lateral drainage structure	910 Ø RCP

Source: Department of Public Works and Highways, Department Order No. 11 series of 2014, Department Order No. 46 dated 25 June 2012, Manila.

Table 3-3 AASHTO Recommended Minimum Width of Traveled Way and Shoulders for Local Rural Roads

Design Speed (kph)	Minimum Width of Traveled Way (m) for Specified Design Volume (vehicles per day)			
	Under 400	400 to 1500	1500 to 2000	Over 2000
20	5.4	6.0 [a]	6.0	6.6
30	5.4	6.0 [a]	6.6	7.2 [b]
40	5.4	6.0 [a]	6.6	7.2 [b]
50	5.4	6.0 [a]	6.6	7.2 [b]
60	5.4	6.0 [a]	6.6	7.2 [b]
70	6.0	6.6	6.6	7.2 [b]
80	6.0	6.6	6.6	7.2 [b]
90	6.6	6.6	7.2 [b]	7.2 [b]
100	6.6	6.6	7.2 [b]	7.2 [b]
All speeds	Width of graded shoulder on each side of the road (m)			
	0.6	1.5 [a,c]	1.8	2.4

[a] For roads in mountainous terrain with design volume of 400 to 600 vehicles per day, use 5.4 m traveled way width and 0.6 m shoulder width.
[b] Where the width of the traveled way is shown as 7.2 m, the width may remain at 6.6 m on reconstructed highways where there is no crash pattern suggesting the need for widening.
[c] May be adjusted to achieve a minimum roadway width of 9 m for design speeds greater than 60 kph.

Source: Table 5-5 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-4 AASHTO Recommended Minimum Width of Traveled Way and Shoulders for Rural Collector Roads

Design Speed (kph)	Minimum Width of Traveled Way (m) for Specified Design Volume (vehicles per day[a])			
	Under 400	400 to 1500	1500 to 2000	Over 2000
30	6.0 [b]	6.0	6.6	7.2
40	6.0 [b]	6.0	6.6	7.2
50	6.0 [b]	6.0	6.6	7.2
60	6.0 [b]	6.6	6.6	7.2
70	6.0	6.6	6.6	7.2
80	6.0	6.6	6.6	7.2
90	6.6	6.6	7.2	7.2
100	6.6	6.6	7.2	7.2
All speeds	Width of graded shoulder on each side of the road (m)			
	0.6	1.5 [c]	1.8	2.4

[a] On roadways to be reconstructed, a 6.6 m traveled way may be retained where the alignment is satisfactory and there is no crash pattern suggesting the need for widening.
[b] A 4.5 m minimum width may be used for roadways with design volumes under 250 vehicles per day.
[c] Shoulder width may be reduced for design speeds greater than 50 kph provided that a minimum roadway width of 9 m is maintained.

Source: Table 6-5 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-5 AASHTO Recommended Minimum Width of Traveled Way and Shoulders for Rural Arterial Roads

Design Speed (kph)	Minimum Width of Traveled Way (m)[a] for Specified Design Volume (vehicles per day)			
	Under 400	400 to 1500	1500 to 2000	Over 2000
60	6.6	6.6	6.6	7.2
70	6.6	6.6	6.6	7.2
80	6.6	6.6	7.2	7.2
90	6.6	6.6	7.2	7.2
100	7.2	7.2	7.2	7.2
110	7.2	7.2	7.2	7.2
120	7.2	7.2	7.2	7.2
130	7.2	7.2	7.2	7.2
All speeds	Width of usable shoulder (m)[b]			
	1.2	1.8	1.8	2.4

[a] On roadways to be reconstructed, an existing 6.6 m traveled way may be retained where the alignment is satisfactory and there is no crash pattern suggesting the need for widening.
[b] Preferably, usable shoulder on arterials should be paved; however, where volumes are low or a narrow section is needed to reduce construction impacts, the paved shoulder width may be a minimum of 0.6 m provided that bicycle use is not intended to be accommodated on the shoulder.

Source: Table 7-3 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

3.3 General Principles for Geometric Design

In preparing the design of a new highway or a redesign of an old one, the designer must give attention to the following basic considerations aimed at functionality, homogeneity and predictability:

- The design must be suitable for the traffic volume, both daily and at the design peak hour, for the design speed and for the character of the vehicles to use the facility.
- The design must be consistent and must avoid surprise changes in alignment, grade and sight distance.
- The design must be pleasing to the user and to those who live along it.
- The design must be complete. However, for the designer to be able to ensure the effectiveness of his design to a large degree, the necessary roadside treatment and the provision of control devices, such as lane markers and special signs, are taken into account.
- The design shall be as simple as possible from the standpoint of the builder. Excessive changes in cross sectional design or the use of a variety of types within a project will in many cases increase the cost and difficulty of construction beyond the commensurate value of such 'uniqueness'.
- The design should be such that the finished road can be maintained at the least cost.
- The design must be safe for driving and should ensure confidence for motorists.

Highway design is also sometimes required to consider staged construction, requiring careful consideration of various aspects including safety; for example, the safety challenges of designing a dual carriageway road with grade separated interchanges that are to operate with only one carriageway initially.

3.4 Design Speed

With reference to Section 2.6.3, design speed is not a major factor for local urban streets and collector streets because their typical grid layout and closely spaced intersections usually limit vehicular speed. Design speeds ranging from 30 to 50 kph for local streets, and 50 kph or higher for collector streets, are normally used in design, depending on available right-of-way, terrain, likely pedestrian presence, adjacent development, and other controls.

Minimum design speeds for Local Rural Roads and Rural Collector Roads are given in Table 3-6 and Table 3-7. In the Philippines it is important to stress that where rural roads pass through villages the MAXIMUM design speed should be that which will reduce the risk to pedestrians to an acceptable level – either 50 kph or 30 kph depending on the volume of pedestrian traffic.

Rural arterials other than expressways should be designed for speeds of 60 to 120 kph, depending on terrain, driver expectancy, and, in the case of reconstruction projects, the alignment of the existing facility.

Urban arterials should be designed for speeds of 50 to 100 kph, with lower speeds in business districts and developed areas, and higher speeds in outlying suburban and developing areas.

Expressways should be designed for speeds not less than 80 kph.

Table 3-6 Minimum Recommended Design Speeds for Local Rural Roads

Type of Terrain	Design Speed (kph) for Specified Design Volume (vehicles per day)					
	Under 50	50 to 250	250 to 400	400 to 1500	1500 to 2000	2000 and over
Level	50	50	60	80	80	80
Rolling	30	50	50	60	60	60
Mountainous	30	30	30	50	50	50

Source: Table 5-1 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-7 Minimum Recommended Design Speeds for Rural Collector Roads

Type of Terrain	Design Speed (kph) for Specified Design Volume (vehicles per day)		
	0 to400	400 to 2000	Over 2000
Level	60	80	100
Rolling	50	60	80
Mountainous	30	50	60

Note: Where practical, design speeds higher than those shown should be considered.

Source: Table 6-1 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

3.5 Road Classification

With reference to Section 2.6.6, local urban streets are usually designed with a standard two-lane cross section, but a four-lane cross section may be appropriate in certain urban areas, as governed by traffic volume, administrative policy, or other community considerations. It is usually difficult and costly to modify the geometric design of an existing street unless provision has been made at the time of initial construction.

Traffic volume is a major factor for streets serving industrial or commercial areas, and urban collector roads. In this case the projected ADT should be the design basis. Collector roads and streets should have a sufficient number of lanes to accommodate design traffic volumes for the desired level of service.

Rural highways usually consist of two-lane local roads, which should be designed to accommodate the highest practical criteria compatible with traffic and topography.

New rural arterial roads, or improvement of existing rural arterial roads, require the determination of the design traffic volume. It is usually appropriate to design high-volume rural arterials using the 30 HV, which is typically about 15% of the ADT on rural roads. The number of lanes on an arterial roadway is determined based on consideration of volume, level of service, and capacity conditions. Multilane arterials may be undivided or divided, depending upon traffic volume and safety considerations.

For expressways, specific capacity needs should be determined from directional design hourly volumes (DDHV) for the appropriate design period. In large metropolitan areas, the selection of appropriate design traffic volumes and design periods may be influenced by system planning. Segments of expressway may be constructed or reconstructed to be commensurate with either intermediate traffic demands or with traffic based on the completed system, whichever may be more appropriate. Rural expressways are generally designed for high-volume and high-speed operation.

3.6 Basic Design Consideration

This section deals with the fundamental considerations in highway design, including reference to the safety concept of the Clear Zone.

The calculated minimum stopping sight distance for various assumed speeds is developed in Table 3-8, while Table 3-9 provides minimum stopping sight distances on grades.

For purpose of design, wet conditions govern in determining stopping sight distances due to the lower coefficients of friction on wet pavements compared to dry pavements.

Table 3-8 Stopping Sight Distance on Level Roadways

Design Speed (kph)	Brake Reaction Distance (m)	Braking Distance on Level (m)	Stopping Sight Distance	
			Calculated (m)	Design (m)
20	13.9	4.6	18.5	20
30	20.9	10.3	31.2	35
40	27.8	18.4	46.2	50
50	34.8	28.7	63.5	65
60	41.7	41.3	83.0	85
70	48.7	56.2	104.9	105
80	55.6	73.4	129.0	130
90	62.6	92.9	155.5	160
100	69.5	114.7	184.2	185
110	76.5	138.8	215.3	220
120	83.4	165.2	248.2	250
130	90.4	193.8	284.2	285

Note: Break reaction distance predicated on a time of 2.5 s; deceleration rate of 3.4 m/s^2 used to determine calculated sight distance.

Source: Table 3-1 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-9 Stopping Sight Distance on Grades

Design Speed (kph)	Downgrades			Upgrades		
	3%	6%	9%	3%	6%	9%
20	20	20	20	19	18	18
30	32	35	35	31	30	29
40	50	50	53	45	44	43
50	66	70	74	61	59	58
60	87	92	97	80	77	75
70	110	116	124	100	97	93
80	136	144	154	123	118	114
90	164	174	187	148	141	136
100	194	207	223	174	167	160
110	227	243	262	203	194	186
120	263	281	304	234	223	214
130	302	323	350	267	254	243

Source: Table 3-2 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

- Distance travelled while the passing vehicle occupies the left lane.
- Distance between the passing vehicle at the end of its maneuver and the opposing vehicle.
- Distance traversed by an opposing vehicle for two-thirds of the time the passing vehicle occupies the left lane, or 2/3 of the distance travelled while the passing vehicle occupies the left lane.

Appreciable grades increase the sight distance required for safe passing. The sight distance required to permit a vehicle travelling upgrade to pass with safety is greater than that required on a level road. This is due to reduced acceleration of the passing vehicle, which increases the time of passing, and due to the likelihood of opposing traffic speeding up increasing the distance travelled by it. For passing to be performed safely on upgrades, the passing sight distance should be greater than the minimum. The designer should recognize the desirability of increasing the minimum shown in Table 3-11, which is sufficient for a single or isolated pass only.

Table 3-11 Passing Sight Distance for Design of Two-Lane Highways

Design Speed (kph)	Assumed Speeds (kph)		Passing Sight Distance (m)
	Passed Vehicle	Passing Vehicle	
30	11	30	120
40	21	40	140
50	31	50	160
60	41	60	180
70	51	70	210
80	61	80	245
90	71	90	280
100	81	100	320
110	91	110	355
120	101	120	395
130	111	130	440

Source: Table 3-4 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

3.6.1.4 Headlight Sight Distance

For night driving on highways without lighting, the length of visible roadway is that roadway that is directly illuminated by the headlights of the vehicle. For certain conditions the minimum stopping sight distance values used for design can exceed the length of visible roadway.

First, vehicle headlights have limitations on the distance over which they can project the light intensity levels needed for visibility, particularly on low beam. Second, sight distance is limited where there are horizontal and vertical alignment curves.

- Ending a curve on a bridge is undesirable, unsightly and adds needless complications to design and construction. Likewise curves beginning or ending near a bridge should be placed such that no part of the superelevation transition extends on to the bridge. Compound curves on a bridge are equally undesirable. If curvature is unavoidable, the bridge should be entirely on a simple curve as flat as physical conditions permit.

3.6.3 Vertical Alignment

Highways should be designed to encourage uniform operation throughout. Use of a selected design speed is a means toward this end by the correlation of various geometric features of the highway. Design values have been determined and agreed upon for many highway features but few conclusions have been reached on roadway grades in relation to design speed. Vehicle operating characteristics on grades and established relationships of grades and their lengths to design speed are discussed in this section.

3.6.3.1 Gradient

For economy of vehicle operation, particularly truck operation, grades should be as flat as possible. However, flat grades in rolling or hilly country require very costly earthwork, or much greater distance, or both. For Philippine national roads, the maximum grade is reduced or 'compensated' on sharp curves according to the following rule: 'For every degree of curvature over 6 degrees, the ruling grade shall be reduced by 0.1%'. On tangents or curves of less than 1 degree, 7% grades are permitted for distances not exceeding one kilometer. Grades exceeding the above limits shall not be used in any national road without permission from the Secretary of the Department of Public Works & Highways and the Director of the Bureau of Design (BOD), DPWH.

The rolling of grades to avoid heavy earthwork is permissible if proper sight distances are preserved. Secondary dips in the profile in which a vehicle may be hidden from view should be avoided. On a long climb it is well to avoid 'adverse grades', i.e. grades which cause a loss of elevation, although there are times when such grades are justified to secure better alignment and reduce excavation when crossing a small gully. If possible place grades so that excavation balances the embankment. In areas subject to inundation grades should be established 0.50 m above maximum water level. Grades should at least be above pipe culverts by 0.60 m. Grades of bridges should allow 1.50 m free board above the maximum flood water elevations to the bottom of girders for streams carrying debris and 1.00 m for others.

Maximum Grades

Maximum grades of 5% are considered appropriate for a design speed of 110 kph. For a design speed of 50 kph maximum grades are generally in the range of 7 to 12%, depending on topography. In the Philippines the maximum grade widely used is 6%.

AASHTO maximum recommended grades for local rural roads, rural collector roads, urban collector roads, rural and urban arterials, and rural and urban expressways of varying design speed are provided in Table 3-27 to Table 3-32. Grades for local residential streets should be as level as practical, consistent with the surrounding terrain, with a minimum grade for streets with outer curbs of 0.30%. Streets in commercial and industrial areas should have grades less than 8%.

Minimum Grades

On through cut sections grades should be at least 0.50% to provide longitudinal drainage. On curved pavements a minimum of 0.30% may be used on high type pavements and accurately crowned to facilitate drainage. Flat or level grades may be used on uncurbed highways which have adequate crown for lateral drainage on high fills.

Table 3-27 AASHTO Recommended Maximum Grades for Local Rural Roads

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)								
	20	30	40	50	60	70	80	90	100
Level	9	8	7	7	7	7	6	6	5
Rolling	12	11	11	10	10	9	8	7	6
Mountainous	17	16	15	14	13	12	10	10	-

Source: Table 5-2 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-28 AASHTO Recommended Maximum Grades for Rural Collector Roads

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)							
	30	40	50	60	70	80	90	100
Level	7	7	7	7	7	6	6	5
Rolling	10	10	9	8	8	7	7	6
Mountainous	12	11	10	10	10	9	9	8

Note: Short lengths of grade in rural areas, such as grades less than 150 m in length, one-way down-grades, and grades on low-volume rural collectors may be up to 2% steeper than the grades shown above.

Source: Table 6-2 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-29 AASHTO Recommended Maximum Grades for Urban Collector Roads

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)							
	30	40	50	60	70	80	90	100
Level	9	9	9	9	8	7	7	6
Rolling	12	12	11	10	9	8	8	7
Mountainous	14	13	12	12	11	10	10	9

Note: Short lengths of grade in urban areas, such as grades less than 150 m in length, one-way down-grades, and grades on low-volume rural collectors may be up to 2% steeper than the grades shown above.

Source: Table 6-8 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-30 AASHTO Recommended Maximum Grades for Rural Arterial Roads

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)							
	60	70	80	90	100	110	120	130
Level	5	5	4	4	3	3	3	3
Rolling	6	6	5	5	4	4	4	4
Mountainous	8	7	7	6	6	5	5	5

Source: Table 7-2 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-31 AASHTO Recommended Maximum Grades for Urban Arterials

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)					
	50	60	70	80	90	100
Level	8	7	6	6	5	5
Rolling	9	8	7	7	6	6
Mountainous	11	10	9	9	8	8

Source: Table 7-4 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-32 AASHTO Recommended Maximum Grades for Rural and Urban Expressways

Type of Terrain	Maximum Grade (%) for Specific Design Speed (kph)					
	80	90	100	110	120	130
Level	4	4	3	3	3	3
Rolling	5	5	4	4	4	4
Mountainous	6	6	6	5	-	-

Source: Table 8-1 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Critical Lengths of Grade for Design

This is the maximum length upgrade wherein a loaded truck can negotiate without unreasonable reduction in speed. This critical grade length is determined from a loaded truck which will effect a reduction in speed of 25 kph below the average running speed. The critical lengths of upgrades in Table 3-33 when approached by a level section should not be used as a control but should be referred to as a guide.

Table 3-33 Critical Lengths of Grade

Critical Length of Upgrade (m)	Upgrade (%)
500	3
340	4
240	5
200	6
170	7
150	8

Table 1-1 Design Vehicle Dimension

Design Vehicle Type	Symbol	Dimensions (m)											
		Overall			Overhang		WB1	WB2	S	T	WB3	WB4	Typical Kingpin to Center of rear Tandem Axle
		Height	Width	Length	Front	Rear							
Passenger Car	P	1.30	2.13	5.79	0.91	1.52	3.35	-	-	-	-	-	-
Single-Unit Truck	SU-9	3.35-4.11	2.44	9.14	1.22	1.83	6.10	-	-	-	-	-	-
Single-Unit Truck (three-axle)	SU-12	3.35–4.11	2.44	12.04	1.22	3.20	7.62	-	-	-	-	-	-
Buses													
Intercity Bus (Motor Coaches)	BUS-12	3.66	2.59	12.36	1.93	2.73 [a]	7.70	-	-	-	-	-	-
	BUS-14	3.66	2.59	13.86	1.89	2.73 [b]	8.69	-	-	-	-	-	-
City Transit Bus	CITY-BUS	3.20	2.59	12.19	2.13	2.44	7.62	-	-	-	-	-	-
Conventional School Bus (65 pass.)	S-BUS 11	3.20	2.44	10.91	0.79	3.66	6.49	-	-	-	-	-	-
Large School Bus (84 pass.)	S-BUS 12	3.20	2.44	12.19	2.13	3.96	6.10	-	-	-	-	-	-
Articulated Bus	A-BUS	3.35	2.59	18.29	2.62	3.05	6.71	5.91	1.89	4.02	-	-	-
Combination Trucks													
Intermediate Semitrailer	WB-12	4.11	2.44	13.87	0.91	1.37 [a]	3.81	7.77	-	-	-	-	7.77
Interstate Semitrailer	WB-19 *	4.11	2.59	21.03	1.22	1.37 [a]	5.94	12.50	-	-	-	-	12.50
Interstate Semitrailer	WB-20 **	4.11	2.59	22.40	1.22	1.37 [a]	5.94	13.87	-	-	-	-	13.87
'Double-Bottom' Semitrailer/Trailer	WB-20D	4.11	2.59	22.04	0.71	0.91	3.35	7.01	0.91 [c]	2.13 [c]	6.86	-	7.01
Double-Semitrailer/Trailer	WB-28D	4.11	2.59	29.67	0.71	0.91	5.33	12.19	1.37	2.13	6.86	-	12.34
Triple-Semitrailer/Trailer	WB-30T	4.11	2.59	31.94	0.71	0.91	3.35	6.86	0.91 [d]	2,13 [d]	6.86	6.86	7.01
Turnpike Double-Semitrailer/Trailer	WB-33D *	4.11	2.59	34.75	0.71	1.37 [a]	3.72	12.19	1.37 [e]	3.05 [e]	12.19	-	12.34
Recreational Vehicles													
Motor Home	MH	3.66	2.44	9.14	1.22	1.83	6.10	-	-	-	-	-	-
Car and Camper Trailer	P/T	3.05	2.44	14.84	0.91	3.66	3.35	-	1.52	5.39	-	-	-
Car and Boat Trailer	P/B	-	2.44	12.80	0.91	2.44	3.35	-	1.52	4.57	-	-	-
Motor Home and Boat Trailer	MH/B	3.66	2.44	16.15	1.22	2.44	6.10	-	1.83	4.57	-	-	-

* Design vehicle with 14.63 m trailer
** Design vehicle with 16.15 m trailer
a This is the length of the overhang from the back axle of the tandem axle assembly
b combined dimension is 5.91 m and articulating section is 1.22 m wide
c Combined dimension is typically 3.05 m
d Combined dimension is typically 3.05 m
e Combined dimension is typically 3.81 m
Source: Table 2-1a in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 1-2 Minimum Turning Radii of Design Vehicles

Design Vehicle Type	Symbol	Minimum Design Turning Radius (m)	Centerline[b] Turning Radius [CTR] (m)	Minimum Inside Radius (m)
Passenger Car	P	7.26	6.40	4.39
Single-Unit Truck	SU-9	12.73	11.58	8.64
Single-Unit Truck (three-axle)	SU-12	15.60	14.46	11.09
Intercity Bus (Motor Coaches)	BUS-12	12.70	11.53	7.41
	BUS-14	13.40	12.25	7.54
City Transit Bus	CITY-BUS	12.80	11.52	7.45
Conventional School Bus (65 pass.)	S-BUS 11	11.75	10.64	7.25
Large School Bus (84 pass.)	S-BUS 12	11.92	10.79	7.71
Articulated Bus	A-BUS	12.00	10.82	6.49
Intermediate Semitrailer	WB-12	12.16	10.97	5.88
Interstate Semitrailer	WB-19 *	13.66	12.50	2.25
Interstate Semitrailer	WB-20 **	13.66	12.50	0.59
'Double-Bottom' Semitrailer/Trailer	WB-20D	13.67	12.47	5.83
Double-Semitrailer/Trailer	WB-28D	24.98	23.77	16.94
Triple-Semitrailer/Trailer	WB-30T	13.67	12.47	2.96
Turnpike Double-Semitrailer/Trailer	WB-33D *	18.25	17.04	4.19
Motor Home	MH	12.11	10.97	7.92
Car and Camper Trailer	P/T	10.03	9.14	5.58
Car and Boat Trailer	P/B	7.26	6.40	2.44
Motor Home and Boat Trailer	MH/B	15.19	14.02	10.67

*Design vehicle with 14.63 m trailer **Design vehicle with 16.15 m trailer

[a] School buses are manufactured from 42-passenger to 84-passenger sizes. This corresponds to wheelbase lengths of 3.35 to 6.10 m respectively. For these different sizes, the minimum design turning radii vary from 8.58 to 11.92 m and the minimum inside radii vary from 5.38 to 7.1 m.

[b] The turning radius assumed by a designer when investigating possible turning paths is set at the centerline of the front axle of a vehicle. If the minimum turning path is assumed, the CTR approximately equals the minimum design turning radius minus one-half the front width of the vehicle.

Source: Table 2-2a in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-20 Adjustment Factor for Number of Lanes Rotated

Number of Lanes Rotated(n_1)	Adjustment Factor* (b_w)	Length Increase Relative to One-Lane Rotated (= n_1 x b_w)
1	1.00	1.00
1.5	0.83	1.25
2	0.75	1.50
2.5	0.70	1.75
3	0.67	2.00
3.5	0.64	2.25

'* bw = [1 + 0.5 (n1 – 1)] / n1

Source: Table 3-16 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-21 Runoff Locations that Minimize the Vehicle's Lateral Motion

Design Speed (kph)	Portion of Runoff Located Prior to the Curve			
	Number of Lanes Rotated			
	1.0	1.5	2.0-2.5	3.0-3.5
20 – 70	0.80	0.85	0.90	0.90
80 – 130	0.70	0.75	0.80	0.85

Source: Table 3-18 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

3.6.2.4 Widening

Due to the fact that on curves the rear wheels of motor vehicles do not ordinarily travel in the same radius as the front wheels, it is desirable to widen the roadbed especially along sharp curves. On simple curves, widening should be applied on the inside edge of pavement only. On curves designed with spirals, widening may be placed on the inside or divided equally between the inside and outside of the curve. The amount of widening required for curves of various radii is shown in Table 3-22.

The minimum widening recommended on curves is 0.60 m and depends on the degree of curvature, design speed and design vehicle. Adjustments for design vehicles other than WB-19 are presented in Table 3-23.

Widening is transitioned within the length of the superelevated runoff applied at the inside edge or on both edges and should be attained linearly.

Radius of Curve (m)	Roadway Width = 7.2 m						Roadway Width = 6.6 m						Roadway Width = 6.0 m					
	Design Speed (kph)						Design Speed (kph)						Design Speed (kph)					
	50	60	70	80	90	100	50	60	70	80	90	100	50	60	70	80	90	100
3000	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.3	0.3	0.3	0.3	0.5	0.6	0.6	0.6	0.6	0.6
2500	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.3	0.3	0.3	0.3	0.4	0.6	0.6	0.6	0.6	0.6	0.7
2000	0.0	0.0	0.0	0.1	0.1	0.1	0.3	0.3	0.3	0.4	0.4	0.4	0.6	0.6	0.6	0.7	0.7	0.7
1500	0.0	0.1	0.1	0.1	0.1	0.2	0.3	0.4	0.4	0.4	0.4	0.5	0.6	0.7	0.7	0.7	0.7	0.8
1000	0.1	0.2	0.2	0.2	0.3	0.3	0.4	0.5	0.5	0.5	0.6	0.6	0.7	0.8	0.8	0.8	0.9	0.9
900	0.2	0.2	0.2	0.3	0.3	0.3	0.5	0.5	0.5	0.6	0.6	0.6	0.8	0.8	0.8	0.9	0.9	0.9
800	0.2	0.2	0.3	0.3	0.3	0.4	0.5	0.5	0.6	0.6	0.6	0.7	0.8	0.8	0.9	0.9	0.9	1.0
700	0.3	0.3	0.3	0.4	0.4	0.4	0.6	0.6	0.6	0.7	0.7	0.7	0.9	0.9	0.9	1.0	1.0	1.0
600	0.3	0.4	0.4	0.4	0.5	0.5	0.6	0.7	0.7	0.7	0.8	0.8	0.9	1.0	1.0	1.0	1.1	1.1
500	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2
400	0.5	0.6	0.6	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3	1.3	1.4
300	0.7	0.8	0.8	0.9	1.0	1.0	1.0	1.1	1.1	1.2	1.3	1.3	1.3	1.4	1.4	1.5	1.6	1.6
250	0.9	1.0	1.0	1.1	1.1		1.2	1.3	1.3	1.4	1.4		1.5	1.6	1.6	1.7	1.7	
200	1.1	1.2	1.3	1.3			1.4	1.5	1.6	1.6			1.7	1.8	1.9	1.9		
150	1.5	1.6	1.7	1.8			1.8	1.9	2.0	2.1			2.1	2.2	2.3	2.4		
140	1.6	1.7					1.9	2.0					2.2	2.3				
130	1.8	1.8					2.1	2.1					2.4	2.4				
120	1.9	2.0					2.2	2.3					2.5	2.6				
110	2.1	2.2					2.4	2.5					2.7	2.8				
100	2.3	2.4					2.6	2.7					2.9	3.0				
90	2.5						2.8						3.1					
80	2.8						3.1						3.4					
70	3.2						3.5						3.8					

Notes: Values shown are for WB-19 design vehicle and represent widening in meters; for other design vehicles use adjustments in Table 3-23
Values less than 0.6 m may be disregarded.
For 3-lane roadways, multiply above values by 1.5.
For 4-lane roadways, multiply above values by 2.
Source: Table 3-26a in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

Table 3-23 Adjustments for Travelled Way Widening Values on Open Highway Curves (Two-Lane Highways, One-Way or Two-Way)

Radius of Curve (m)	Design Vehicle							
	SU-9	SU-12	WB-12	WB-20	WB-20D	WB-28D	WB-30T	WB-33D
3000	-0.4	-0.3	-0.3	0.0	0.0	0.0	0.0	0.0
2500	-0.4	-0.4	-0.3	0.0	0.0	0.0	0.0	0.1
2000	-0.4	-0.4	-0.4	0.0	0.0	0.0	0.0	0.1
1500	-0.4	-0.4	-0.4	0.0	-0.1	0.0	0.0	0.1
1000	-0.5	-0.4	-0.4	0.0	-0.1	0.0	0.0	0.1
900	-0.5	-0.4	-0.4	0.0	-0.1	0.0	0.0	0.1
800	-0.5	-0.5	-0.4	0.0	-0.1	0.0	0.0	0.2
700	-0.5	-0.5	-0.5	0.1	-0.1	0.1	0.0	0.2
600	-0.6	-0.5	-0.5	0.1	-0.1	0.1	-0.1	0.2
500	-0.6	-0.6	-0.5	0.1	-0.2	0.1	-0.1	0.3
400	-0.7	-0.6	-0.6	0.1	-0.2	0.1	-0.1	0.3
300	-0.8	-0.7	-0.7	0.1	-0.3	0.1	-0.1	0.4
250	-0.9	-0.8	-0.8	0.1	-0.3	0.2	-0.1	0.5
200	-1.1	-1.0	-0.9	0.2	-0.4	0.2	-0.2	0.6
150	-1.3	-1.2	-1.1	0.2	-0.6	0.3	-0.2	0.8
140	-1.4	-1.2	-1.2	0.3	-0.6	0.3	-0.2	0.9
130	-1.5	-1.3	-1.2	0.3	-0.6	0.3	-0.2	1.0
120	-1.6	-1.4	-1.3	0.3	-0.7	0.3	-0.3	1.1
110	-1.7	-1.5	-1.4	0.3	-0.8	0.4	-0.3	1.2
100	-1.8	-1.6	-1.5	0.4	-0.8	0.4	-0.3	1.3
90	-2.0	-1.8	-1.6	0.4	-0.9	0.4	-0.4	1.4
80	-2.2	-1.9	-1.8	0.5	-1.0	0.5	-0.4	1.6
70	-2.5	-2.2	-2.0	0.5	-1.2	0.6	-0.5	1.9

Notes:
Adjustments are applied by adding to or subtracting from the values in Table 3-22.
Adjustments depend only on radius and design vehicle; they are independent of roadway width and design speed.
For 3-lane roadways, multiply above values by 1.5.
For 4-lane roadways, multiply above values by 2.

Source: Table 3-27 in AASHTO, 2011, A Policy on Geometric Design of Highways and Streets 6th Edition. Used by Permission.

3.6.2.5 Spiral Curve Transition

If a vehicle is running at high speed on a carefully restricted path made up of tangents connected by sharp circular curves, riding is extremely uncomfortable. As the car approaches a curve, superelevation begins and the vehicle is titled inward, but the passenger must remain vertical since there is no centrifugal force requiring compensation. When the vehicle reaches the curve full centrifugal force develops at once, and pulls the rider outward from his position. This process is repeated in reverse order as the vehicle leaves the curve. When spiral curves are introduced, the change in radius from infinity on the tangent to that of the circular curve is effected gradually so that centrifugal force also develops similarly.